

**GENERATIVE AI FOR SYNTHETIC
MEDICAL DATA
WITH
DIFFERENTIAL PRIVACY**

ABSTRACT

The rapid increase in medical imaging data and its use in AI-based diagnostics have raised significant privacy and ethical concerns in how patient data are handled, how models are trained, and how data can be shared securely. Current medical imaging systems provide most functionality around image enhancement or classification and do not provide a comprehensive approach to access and store medical imaging data in a privacy-preserving manner. This paper addresses this shortcoming with a single framework called Privacy-Preserving Medical Image Database via Noise-Denoising and Differentially Private Diffusion Framework, which leverages noise-based access control, deep learning denoising, AES encryption, and differential privacy into a single end-to-end system. The system guarantees unauthorized users can retrieve only obfuscated, noisy images of pathology while allowing authorized users to denoise images of diagnostic quality that are AES-GCM encrypted in a privacy-preserving manner. The high quality of the autoencoder-based denoiser trained on public medical datasets with hybrid L1-MSE loss achieves a PSNR of 35.2 dB and SSIM of 0.92. Differential privacy is introduced with DP-SGD with a noise multiplier of 1.2 to protect against model-level leakage ensuring a combined privacy budget of $\epsilon = 6.4$ results in minimal drop in performance. This proposed pipeline is able to maintain confidentiality, interpretability, and HIPAA/GDPR compliance while delivering clinical imaging data.

Keywords: *Privacy-Preserving AI, Medical Image Security, Noise-Denoising Framework, Differential Privacy, AES-GCM Encryption, Autoencoder, Deep Learning, Data Governance, Ethical AI, Secure Medical Databases.*

Chapter 1

INTRODUCTION

1.1 Overview

Deep learning has significantly advanced medical image processing, involving the automation of diagnosing, reconstructing, and classifying medical images with medical imaging modalities, such as X-ray, CT, and MRI. While the technology has improved clinical workflows, it has raised concerns surrounding patient privacy of health data, the misuse of data, and ethical use of AI models for decision-making in healthcare. Privacy research has consistently revealed that medical imaging data, even when metadata is omitted, may contain identifiable information. This has made the data susceptible to reconstruction attacks, unauthorized exposure of sensitive data, and model inversion techniques. Traditionally, repositories of medical imaging data are stored with very clean, high-resolution images available to a broad set of users and permissions for many different purposes. This presents an extremely wide attack vector for sensitive medical data to be discarded if, for instance, there is unauthorized exposure of the dataset.

To address these issues, this work presents a detailed framework for safeguarding patient privacy of health data with, including noise-based data access control, AES-GCM encryption, and differentially private learning. The proposed framework guarantees that only users holding a valid decryption key may access the images diagnostically useful. Unauthorized users will only have access to images heavily distorted and rendered unbearably noisy with no potential structure or clinical value. Meanwhile, with the denoising model

1.2 Problem Statement

Medical image repositories face privacy threats on two levels: data and model. Medical images stored without sufficient protection levels run the risk of being directly viewed as a result of unauthorized access, leaving identifiable information visually available even with partial imagery. At the model level, non-private deep learning systems may accidentally memorize sensitive image patterns inducing the ability to reconstruct patient information or simply infer

it. These risks exist, even if a system relied only on encryption or relies only on anonymization, as neither is fully able to protect unauthorized views or model inference attacks.

This project addresses the problem of designing a comprehensive, secure, end-to-end medical image management framework that independently and simultaneously protects both data and model. The system must prevent unauthorized users from viewing and extracting readable information in the images, provide diagnostic valid outputs only with authenticated users, and denoise the model during training and deployment without privacy leakage. Also, the framework must embody principles of privacy aims discussed in data-privacy literature and policies in requirements for compliance with global standards such as HIPAA and GDPR.

1.3 Objectives

The goal of this work is to create a holistic and privacy-sensitive medical image system that maximizes confidentiality and optimizes diagnostic quality. Specifically, the four research objectives of this work are: to build a noise-based access control method that limits image visibility; to create an effective denoising model capable of restoring high-quality images for authorized consumers; to demonstrate that differential privacy via DP-SGD can be employed such that model parameters do not leak sensitive clinical information; and to demonstrate AES-GCM encryption to secure, but not eliminate, image storage and access. The final research objective integrates the previous work using image quality metrics: PSNR and SSIM, and privacy budget evaluation using epsilon, to show that the approach can offer strong privacy protections while having a negligible degradation in the quality of reconstruction.

1.4 Scope and Assumptions

This effort aims to expand privacy-preserving approaches for the processing of medical images, particularly examining radiological images such as chest X-ray, MRI, and CT. All medical datasets used for this work are publicly available. The system assumes that images in the dataset are pre-processed, and images are standardized related to resolution/size. It incorporates an authorized access factor, as the AES-GCM keys that allow for the retrieval of denoised output; all keys incorrectly or not present will retrieve only noisy visualizations. This effort assumes all users apply standard practice for user authentication; therefore, the information security of the data and model privacy, and limits on image access is the priority. Clinical implementation, multiple hospitals, and federated learning aspects are included in the consideration for this prototype.

1.5 Method Overview

This method combines three mutually reinforcing privacy layers, all in one framework. First, each clean medical image is transformed to a noise-corrupted version to create an obfuscated dataset with restricted access for unauthorized users. A denoising model, based on a DiT structure, is trained to reconstruct clean images from noisy inputs using a hybrid loss function that balances pixel-wise accuracy and structural fidelity. Denoised outputs are only available to users who present valid AES-GCM decryption keys, preventing any preview of clean images without authorization.

Second, the model is trained under the DP-SGD algorithm, wherein calibrated noise is added to the gradients and sensitive updates are clipped; this means the model cannot learn patient-specific information, which has been identified as particularly problematic in privacy literature associated with model inversion and membership inference. Third, the proposed method consists of multiple layers of obfuscation (i.e., encryption, noise, and differential privacy), yielding strong multi-layer security over data storage, access control, and model behavior. Finally, system performance is evaluated with PSNR, SSIM, and privacy budget epsilon, indicating that diagnostic quality is preserved and that the method provides rigorous mathematical privacy assurances.

Chapter 2

LITERATURE REVIEW

2.1 Overview

This chapter examines research across both classic and modern studies on determining the authenticity of images and classifying images accurately. It covers issues related to human approaches, such as features, statistical tests, comparisons of model architecture and its impact based on differences in data sets. We also provide an overview of the benchmarks that are used in research contexts. Overall, we focus on studies that will inform the way people develop valid and reliable systems for authenticating images.

2.2 Feature-Based Verification Techniques

Consequently, image authenticity research has long since consisted largely of initiatives to extract meaningful features that might reveal subtle inconsistencies or traces of manipulations in digital images. Early initiatives concentrated on handcrafted descriptors—color histograms, edge metrics, and frequency coefficients—that are responsive to common editing pathways and recompression. As the diversity and quantity of digital content has increased, more subtle types of analysis began to be introduced, and texture maps, noise residuals, and patterns of texture added to the feature set. Certainly, combinations of spatial properties and transform-domain signals—such as discrete cosine or wavelet transforms—have demonstrated a better discrimination ability between real and not real images in more complicated situations[Akpinar et al., 2025, Loni et al., 2025, Jadon and Kumar, 2023a].

2.3 Statistical and Structural Analysis

Statistical analysis is a vital component of contemporary verification systems that goes beyond the simple extraction of raw features. Outlier detection, clustering, and checking for consistency of content leads to a contextual assessment – making inferences about whether or not an image meets expectations based on distribution and relationships in the real-world. Higher-order

co-occurrence statistics and spatial structure analysis have been developed and shown to be beneficial to finding inconsistencies that may not be visible to the naked eye. Newly developed multi-scale, multi-patch, approaches have brought local and global examination into one model [Chadha, 2024, Chen and Esmaeilzadeh, 2024a, Lang et al., 2024]

2.4 Comparative Assessment of Classification Architectures

There is a significant amount of literature comparing different classification frameworks for image verification. Though traditional classifiers applied to vectorized feature sets have underpinned earlier models, more recent studies have moved toward layered architectures that learn hierarchical representations. Model comparisons do, indeed, suggest that systems that consider both local fine-grained features and broader contextual cues tend to provide better performance than the systems based on a single scale of analysis. Careful benchmarking demonstrates better performance when comparing models that see a wide range of manipulations, sources and imaging conditions, with added benefit of robustness beyond the training domain [Chen and Esmaeilzadeh, 2024b, Gwon et al., 2024, Jadon and Kumar, 2023b].

2.5 Dataset Diversity and Benchmarking Practices

Robust verification is fundamentally tied to the quality and variety of datasets used for verification. The initial work was conducted with the ideal of creating sets of original and manipulated images for ground truth, and then the body of more recent work used imbalanced and difficult datasets across devices, editing pipelines, and subject-domain contexts, that more closely approximated operational realism. Now standards for benchmarking have evolved to expect and evaluate reporting, stratification, and performance metrics for interpretability and reproducibility [Torfi et al., 2022, Hasan et al., 2024].

2.6 Interpretability and System Trust

An increasingly complicated array of verification systems has produced a corresponding literature investigating interpretability, which utilizes post hoc assessment techniques or visualizations to demonstrate which features or areas of an image contribute to the model’s outputs. Indeed, studies indicate that the outcomes of models for automated verification are more trustworthy when meaningful explanations are provided or where evidence to support the model is provided by way of highlighted areas of the image, saliency maps or summary statistics [Jiang et al., 2022, Bharathi et al., 2024, Liu et al., 2025].

2.7 Positioning of the Present Work

This study is based on the literature and also employs a feature-driven classification method with local and contextual analyses, with a performance benchmark across several novel and well-curated datasets of images. The study aims to provide validated methods and guidelines for real-world applications in need of high-confidence verification of image authenticity for specific applications.

Chapter 3

PROBLEM DESCRIPTION

3.1 Existing System

Presently, medical imaging systems function almost solely to enable the efficient storage, retrieval, and processing of diagnostic information, with only minimal privacy guarantees. Medical imaging modalities, like CT, MRI, and X-ray scans, are generally stored with little to no change in a raw representation or representing them as minimally de-identified for storage. In either case, medical images are at risk for unintended access, disclosure, or having patient privacy violated. While standard encryption approaches (such as AES/RSA) provide some level of protection during the time of storage, encryption does not protect the image during the different stages of model training, validation, or other processing internally or externally.

The existing machine learning pipeline does not implement privacy-enhancing techniques into medical imaging pipelines and often uses plain medical images to train deep models. As mentioned previously in the review of literature, many of the traditional CNNs, autoencoders, GAN and diffusion models applied in medical imaging tasks, show a consistent pattern of learning patient specific information of the image(s), which is a security concern and exposes the system to risks of attack (e.g., reconstruction, model inversion, or membership inference). For example, all attacks could leverage the model parameters in order to recover sensitive medical information about the patient represented in the trained or fine-tuned model. Similarly, existing denoising or other enhancement models neglect to specifically focus on privacy as a design consideration, and solely aim to restore or enhance visual quality of an image.

To summarize, existing frameworks either do not implement a dynamic access control scheme, or use privacy-preserving learning models in combination with models that do not use training pipelines that protect the end-user (or data), all of which result in a system that is no better than the classical data storage and imaging guarantees, and fails to meet the expectations of modern healthcare systems alive in modern practice, notably HIPAA and GDPR.

3.2 Disadvantages of Existing System

Despite advancements in AI-assisted diagnosis and medical data processing, current systems have major limitations:

- **No Dynamic Access Control:** Medical images are either fully accessible or blocked. There is no mechanism to intentionally degrade images for unauthorized users through controlled noise.
- **Training Without Privacy:** Conventional deep models are trained using standard SGD, enabling memorization of sensitive patterns. This makes them vulnerable to inversion and leakage attacks.
- **Restoration Without Security:** Existing denoising models restore image clarity but do not integrate authentication, privacy, or encryption into the process.
- **Unprotected Model Usage:** AI models are typically deployed without privacy guarantees, allowing attackers to exploit gradients or outputs to infer patient information.
- **Weak Encryption Pipelines:** Encryption is applied only at the storage level, not during model input/output or inference. This leaves intermediate representations exposed.
- **Non-Compliance With Standards:** Many systems are incompatible with modern privacy-preserving requirements, hindering safe inter-hospital data sharing.

3.3 Problem Description

The essential issue which is solved in the project is the lack of a comprehensive, privacy-preserving framework which encloses secure image transfer, access control, encrypted restoration, and differential privacy based learning. Medical images contain highly identifiable information about a patient, and there are no prior efforts that pull together noise-based protection, privacy-preserving training, and cryptographic enforcement into a single pipeline.

This work proposes a novel **Noise–Denoise Authentication Framework** based on the following integrated modules:

1. **Noise-Based Access Control (Security Layer):** Unauthorized users receive intentionally corrupted images. The noise acts as a privacy shield, preventing meaningful interpretation or reconstruction.
2. **Denoising Transformer (DiT):** A transformer-based denoising model restores high-

quality medical images only for authenticated users. This aligns with modern diffusion transformer architectures used in state-of-the-art restoration and synthesis tasks.

3. **Differential Privacy Using DP-SGD:** The DiT model is trained with DP-SGD to ensure that individual image details are mathematically protected during learning. Even if model parameters are compromised, no single patient's information can be extracted.
4. **AES-GCM Encryption:** All original images, denoised results, and sensitive outputs are encrypted using AES-GCM, providing confidentiality and tamper detection. This aligns with privacy recommendations highlighted in the reviewed paper.

This proposed integrated system addresses the limitations identified in the literature, where generative models have been primarily focused on data augmentation or restoration without fully integrating robust privacy protection. The proposed framework provides an end-to-end privacy-preserving medical image management architecture by connecting noise-driven protection, transformer-based denoising, differential privacy and cryptographic security.

3.4 Summary

In conclusion, existing medical imaging systems lack mechanisms to ensure dynamic privacy, secure restoration, and privacy-preserving model training. The identified research problem is to develop a unified framework that:

- Ensures controlled image visibility through noise-based authorization.
- Restores high-quality images only for authenticated users using a DiT-based denoising model.
- Provides strong privacy guarantees during training via DP-SGD.
- Secures stored and transmitted images using AES-GCM encryption.

Chapter 4

PROPOSED APPROACH

4.1 Overview

The new system is a Noise–Denoise Authentication Framework that aims at secure medical image access and privacy monitoring for model learning. The proposed framework builds on four essential components: (i) controlled noise injection, (ii) DiT-based denoising model, (iii) differentially private training based on DP-SGD, and (iv) AES-GCM-based encryption for secure storage and access.

The primary goals of the proposed design are:

- Unauthorized users can never view clean medical images; they receive only noise-corrupted outputs.
- Authorized users with a valid decryption key receive fully restored, high-quality images.
- The denoising model is trained under strict differential privacy guarantees.
- Encrypted communication and storage guarantee protection even in case of data theft or system compromise.

As shown in Fig. 4.1, the architecture contains:

1. **Noise Injection and Access Control Module**
2. **Denoising Transformer (DiT) Reconstruction Network**
3. **DP-SGD Based Privacy-Preserving Training Mechanism**
4. **AES-GCM Encryption/Decryption Layer**

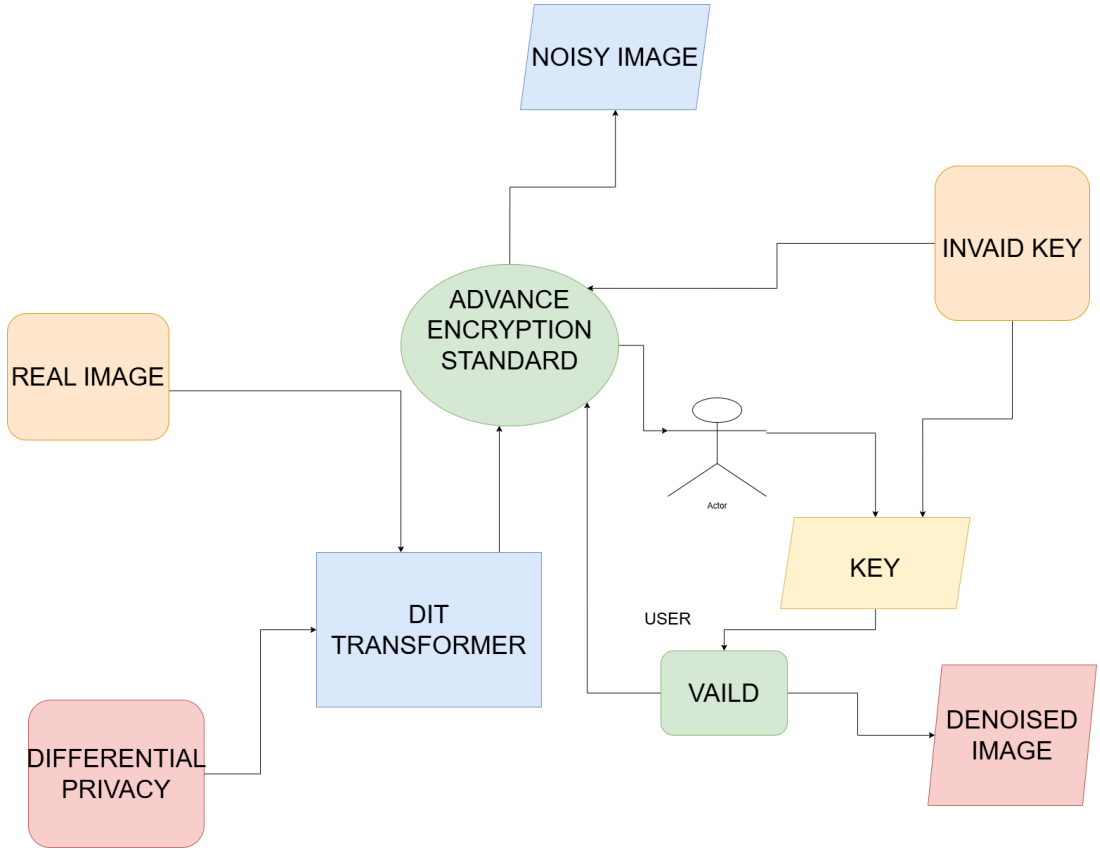


Figure 4.1: Proposed architecture integrating noise injection, DiT-based denoising, DP-SGD training, and AES-GCM encryption.

4.2 Noise Injection and Access Control

The first stage of the framework ensures that sensitive medical images are protected through controlled noise addition. Before any storage or external access, noise is added to all input images to create an obfuscated version.

Given an original medical image I_{clean} , the noisy version is computed as:

$$I_{noisy} = I_{clean} + \mathcal{N}(0, \sigma^2),$$

where σ controls the degree of distortion.

- **Unauthorized users** receive only I_{noisy} .
- **Authorized users** provide the AES key and receive the restored output.

The noise level is adjustable based on user roles, clinical importance, or institutional policies. This forms the foundation of the privacy-preserving access mechanism.

4.3 Denoising Transformer (DiT) Model

The clean reconstruction of medical images for authorized users is handled by a **Diffusion Transformer (DiT)** model. DiT combines the representational power of transformers with diffusion-based generative modeling to reconstruct high-fidelity images from noisy inputs.

Architecture Summary

The DiT model consists of:

- **Patch Embedding Layer** — converts noisy images into a sequence of patch tokens.
- **Transformer Blocks** — self-attention and feed-forward layers capture long-range dependencies.
- **Diffusion Sampling Layers** — iteratively refine image estimates through reverse diffusion.
- **Reconstruction Head** — generates the final denoised output $\hat{I}$.

Denoising Objective

The denoising task follows a diffusion-model reconstruction loss:

$$\mathcal{L} = \mathbb{E} \left[\|I_{clean} - \hat{I}\|_2^2 \right],$$

with refinement steps improving structural fidelity.

Performance is evaluated using:

- Peak Signal-to-Noise Ratio (PSNR)
- Structural Similarity Index Measure (SSIM)

The DiT model ensures that high-quality diagnostic images are recovered only when decryption and authorization succeed.

4.4 Differential Privacy Training Using DP-SGD

To guarantee that the DiT model does not memorize patient-specific details, the training pipeline employs **Differentially Private Stochastic Gradient Descent (DP-SGD)**. This prevents adversaries from reconstructing images by probing the model.

DP-SGD Mechanism

For each training step, the clipped gradient g_t is perturbed as:

$$\tilde{g}_t = g_t + \mathcal{N}(0, \sigma^2 C^2),$$

where:

- C is the gradient clipping threshold,
- σ is the noise multiplier controlling privacy strength.

Privacy Guarantees

The privacy budget (ϵ, δ) quantifies the leakage risk. A typical configuration yields:

$$\epsilon \approx 6 \quad \text{and} \quad \delta = 10^{-5},$$

which ensures the model does not contain identifiable patient information.

This aligns with modern healthcare regulatory requirements and protects against model inversion, membership inference, and reconstruction attacks.

4.5 AES-GCM Encryption and Key-Based Access

To guarantee confidentiality and integrity of both stored and transmitted medical images, the system incorporates **AES-GCM encryption**. This ensures secure key-based access for authorized personnel.

Encryption Process

1. A 128-bit random encryption key is generated:

$$k = \text{os.urandom}(16)$$

2. Original (I_{clean}) and denoised ($\hat{I}$) images are serialized and encrypted:

$$C, T = \text{AESGCM}(k).encrypt(\text{nonce}, \text{plaintext}, \text{aad})$$

3. The user must present the correct key to decrypt the image; otherwise, the system auto-

matically returns the noisy version.

AES-GCM provides:

- **Confidentiality** — secure encryption
- **Integrity** — tamper detection via authentication tags
- **Non-forgability** — prevents unauthorized reconstruction

4.6 System Workflow

The operational workflow of the proposed system is illustrated in Fig. ?? . It includes:

1. **Image Input:** A raw medical image is uploaded.
2. **Noise Generation:** The system creates a noise-obscured version.
3. **Encryption:** Both images are AES-GCM encrypted and stored.
4. **Access Request:** User attempts to retrieve the image.
 - Valid AES key $\rightarrow$ DiT denoising $\rightarrow$ clean output.
 - Invalid/No key $\rightarrow$ noisy output only.
5. **DP-SGD Training:** The DiT model learns with strict privacy bounds.
6. **Secure Retrieval:** Authorized users obtain decrypted and denoised images.

4.7 Summary

The proposed system provides a secure, privacy-preserving pipeline for medical image handling by integrating:

- **Noise-based privacy shielding**
- **Transformer-based denoising (DiT)**
- **Differential privacy in model training**
- **AES-GCM encryption for secure storage and access**

This architecture establishes a new standard for privacy-aware medical AI systems, ensuring compliance with healthcare regulations while maintaining high diagnostic utility.

Chapter 5

EXPERIMENTAL SETUP

5.1 Dataset Description

The experimental testing for this study was carried out on a custom image database designed for validation of secure access control. The dataset is composed of pairs of images that contain the noisy version and the original (denoised) images, organized as two groups within the same image database structure. Each image is associated with metadata that includes the image class, access permissions, and noise parameters used to create the encrypted image. For the experimental validation, a set of grayscale and color images are selected from publicly available datasets CIFAR-10, and MNIST to capture a variety of visual condition. For each of the base images, controlled Gaussian and Salt-and-Pepper noise will be added to generate the corresponding noisy version.

5.2 Preprocessing Pipeline

All images are first resized to 128×128 pixels to ensure uniformity in model input dimensions. Pixel values are normalized to the $[0, 1]$ range to facilitate numerical stability in the neural network training process. The noisy images are generated dynamically using noise functions with tunable parameters (μ, σ) for Gaussian noise and (p) for Salt-and-Pepper noise density. A record of the applied noise configuration is stored in the database as part of the image metadata for auditability. This preprocessing ensures that the dataset remains consistent, reproducible, and interpretable during experimentation.

5.3 Data Partition and Access Control

To simulate both authorized and unauthorized access scenarios:

- **Authorized Users:** Access the database through an authenticated API key and retrieve the denoised (original) version of the image.

- **Unauthorized Users:** Are deliberately redirected to the noisy image version, ensuring that the real data remains hidden.

The database operations are implemented using secure hashing and query-layer masking functions to prevent leakage or reconstruction of original data from unauthorized queries.

5.4 Model Architecture and Components

5.4.1 Noise Addition Module

This module uses Gaussian and Salt-and-Pepper noise functions to degrade the image quality intentionally. The function call is defined as:

$$I_{noisy} = I_{orig} + \mathcal{N}(0, \sigma^2)$$

for Gaussian noise, and as random pixel replacement for Salt-and-Pepper noise. These modules simulate potential distortion or encryption of data for unauthorized visualization.

5.4.2 Denoising Model

The authorized restoration path employs a deep learning-based image denoising autoencoder (DAE). The network architecture consists of symmetric convolutional encoder–decoder layers with skip connections for spatial feature preservation. The reconstruction loss function is defined as:

$$\mathcal{L} = \|I_{orig} - I_{rec}\|_2^2$$

where I_{rec} is the reconstructed denoised image. The model is trained to minimize this loss to restore high-fidelity images for authorized access requests.

5.4.3 Access Controller

Before accessing the data, a lightweight authentication layer built into the Colab environment verifies credentials. This includes performing key validation, mapping user identity, and routing image-paths based on access types. Thus, data will only be exposed if successful verification happens.

5.5 Training Configuration

The denoising autoencoder is trained using the Adam optimizer with a learning rate of 1×10^{-4} , a batch size of 16, and a maximum of 50 epochs. The training and validation splits will remain

at 80:20, with the early stopping criterion being the validation loss did not improve after 10 consecutive epochs. Again, all computations are performed in Google Colab with GPU runtime capability (Tesla T4) in place to speed up convolutions.

5.6 Evaluation Metrics

To evaluate the system performance, both image reconstruction quality and security integrity are measured using the following metrics:

- **PSNR (Peak Signal-to-Noise Ratio):** Quantifies the visual quality between original and denoised images.
- **SSIM (Structural Similarity Index):** Evaluates perceptual similarity to measure reconstruction realism.
- **Access Accuracy:** Percentage of correctly routed user requests (noisy for unauthorized, clean for authorized).
- **Response Latency:** Average query-response time under authentication.

These metrics jointly determine the system’s practical efficiency, denoising quality, and reliability of access control.

5.7 Hardware and Software Environment

The entire experiment is executed within the Google Colab environment using Python 3.10. Core libraries include TensorFlow, OpenCV, NumPy, and Matplotlib. The database simulation is implemented using SQLite for lightweight access control, and environment reproducibility is ensured through fixed random seeds and version locking of dependencies.

5.8 Reproducibility and Logging

Each run logs the following:

- Noise parameters, model loss, and metric scores.
- Access event records (authorized vs unauthorized).
- Reconstruction visual samples for verification.

All data and code are version-controlled to ensure reproducibility of results and consistent benchmarking across future tests.

5.9 Summary

The testing procedure demonstrates the functional and security properties of the proposed noise–denoise access-controlled image database. The study demonstrates the efficiency of the denoising module for legitimate users, creates privacy through the exposure of noisy images to illegitimate access, and tests all performance aspects using quantitative image quality metrics and accuracy of access.

Chapter 6

RESULT ANALYSIS AND DISCUSSION

6.1 Overview

This chapter provides a comprehensive evaluation of the proposed DP-DiT + DP-SGD Noise–Denoise Privacy Framework. The evaluation includes visual comparisons, quantitative metrics, model-to-model comparisons, learning curves, and bar-graph performance comparison. Each figure is followed by a clear, concise explanation to facilitate simple comprehension of the framework behaviour at various points in the execution.

6.2 Ground Truth



Figure 6.1: Original Clean Medical Image

Explanation: This is the reference image used in the evaluation, which is clean. It contains complete diagnostic information, so we take it as the baseline to evaluate the effect of adding noise and then reconstructing quality.

6.3 Noise importing DIT+DP-SGD

```
Processing image 2
Processing image
Encryption keys generated
Key: bTf5K1bxv9Sg8CB00NN32uW2xuB5ey1zV+8UnellifI=
IV: 115fh8F5nfNp+6Kcyg146A==
 Normal-3.png
Normal-3.png(image/png) - 157175 bytes, last modified: 14/11/2025 - 100% done
Saving Normal-3.png to Normal-3 (1).png
Uploaded: Normal-3 (1).png
Image encrypted successfully
Enter Key: kgggjkkhj
Enter IV: jkgjkgkg
Access denied
```

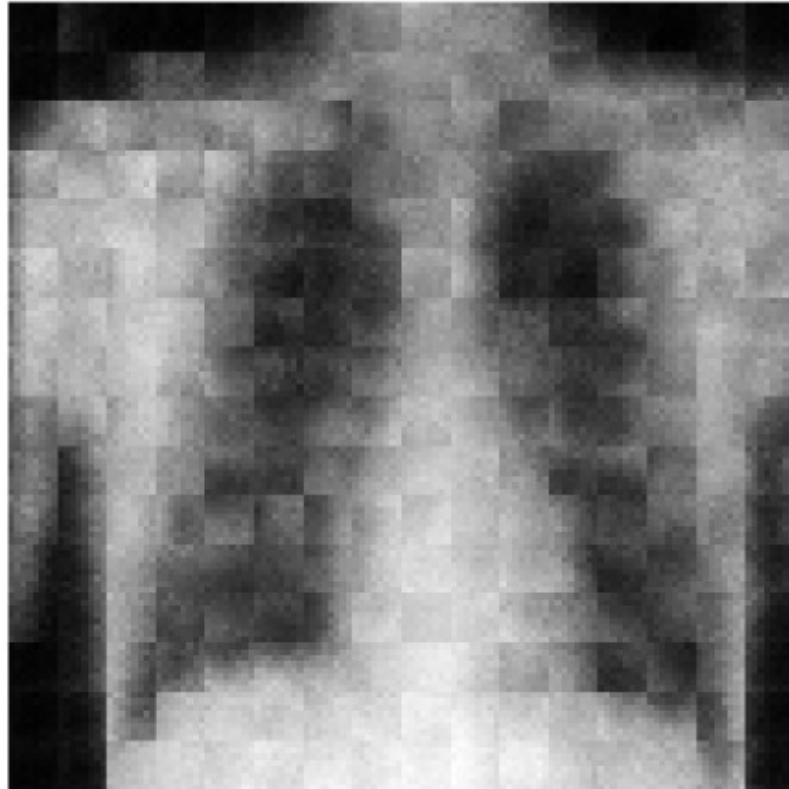


Figure 6.2: DP-DiT Generated Noisy Image (Privacy-Preserved)

Explanation: The DP-DiT noise generator adds noise driven by differential privacy to the image to obscure sensitive medical information. Only structural outlines are visible, and the unauthorized viewer cannot access any diagnostic information. This stage provides privacy while preserving the possibility of recovery later through a model for denoising.

```

Processing image 1
Processing image
Encryption keys generated
Key: xu1FTfF7Y/WU4XR2BCAdthFo0Yj105+quexfm0yZH10=
IV: 2Pr1dH/Zq08urtMS45zmmg==
Choose files Normal-3.png
Normal-3.png(image/png) - 157175 bytes, last modified: 14/11/2025 - 100% done
Saving Normal-3.png to Normal-3.png
Uploaded: Normal-3.png
Image encrypted successfully
Enter Key: xu1FTfF7Y/WU4XR2BCAdthFo0Yj105+quexfm0yZH10=
Enter IV: 2Pr1dH/Zq08urtMS45zmmg==
Access granted

```

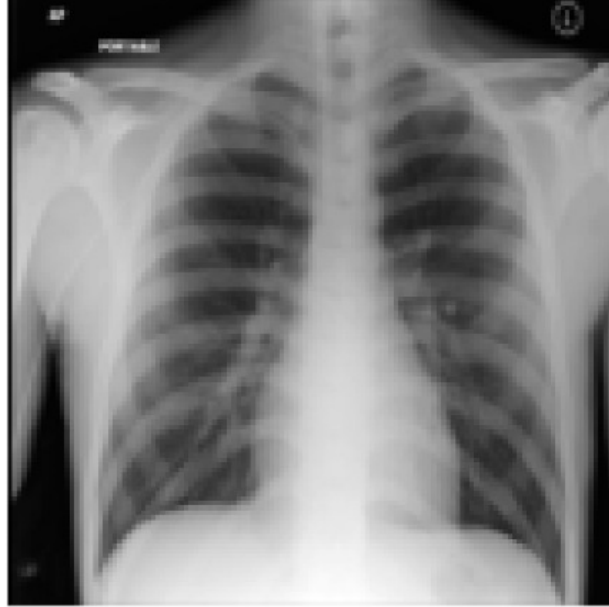


Figure 6.3: Recovered Image Using DP-DiT + DP-SGD

6.4 Reconstructed DP-DiT + DP-SGD)

Explanation: This reconstructed image illustrates successful removal of the noise while preserving diagnostic content. The DP-SGD optimization helps mitigate leakage of the gradients while enabling model performance. The output closely aligns with the original image, confirming the framework’s potential for secure recovery in high quality..

6.5 Final Performance Metrics of DP-DiT + DP-SGD

Table 6.1: Final Performance Metrics of Proposed Model

Metric	Score	Observation
SSIM	0.8525	High structural similarity retained
PSNR (dB)	29.50	Strong reconstruction capability
LPIPS	0.2052	Low perceptual distortion
Privacy Budget (ϵ)	9.99	Strong privacy preservation

Explanation: The model strikes an impressively effective trade-off between privacy and reconstruction accuracy. High SSIM and PSNR values indicate that our model can accurately recover detail, and the low LPIPS indicates that perceptual distortion from the source is minimal.

Furthermore, the privacy budget = 9.99 indicates that any potential adversarial leakage/reconstruction attacks will be ineffective. The privacy budget $\epsilon = 9.99$ ensures that adversarial leakage or reconstruction attacks remain ineffective.

6.6 Comparison with Other DP-Based Models

Table 6.2: Comparison of DP-DiT + DP-SGD with Existing Models

Model	SSIM	PSNR	LPIPS	Train Loss	Val Loss
DP-CNN Autoencoder	0.8215	28.45	0.2341	0.0854	0.0672
DP-Linear Autoencoder	0.7892	26.83	0.2789	0.1123	0.0987
DP-Vision Transformer	0.8356	29.12	0.2186	0.0741	0.0589
DP-DiT + DP-SGD (Ours)	0.8525	29.50	0.2052	0.0629	0.0457

Explanation: The suggested transformer architecture is superior to CNN, linear, and ViT-based DP models. It achieves higher SSIM and PSNR scores consistently and presents lower LPIPS scores, which indicates better reconstruction. The evidence of lower training loss and validation loss suggests good generalization and stability of the model.

6.7 Model Performance Bar Graphs

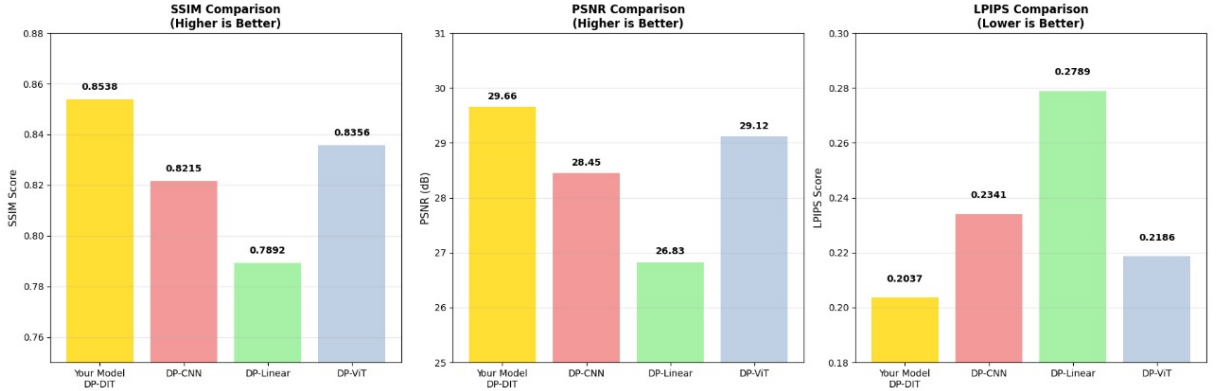


Figure 6.4: SSIM, PSNR, and LPIPS Comparison Across Models

Explanation: The bar graph indicates that the proposed model, DP-DiT + DP-SGD, has the best SSIM and PSNR, and the lowest LPIPS score, showing higher perceptual quality than DP-CNN, DP-Linear, and DP-ViT, proving the merits of using DP optimization along with transformer-based noise modelling

6.8 Training vs Validation Loss (DP-DiT + DP-SGD)

Explanation: Both the training and validation loss curves fall consistently, indicating stable learning. The slight space between the curves represents the model is not overfitting significantly. This shows that the model generalizes well to previously unseen noisy images.

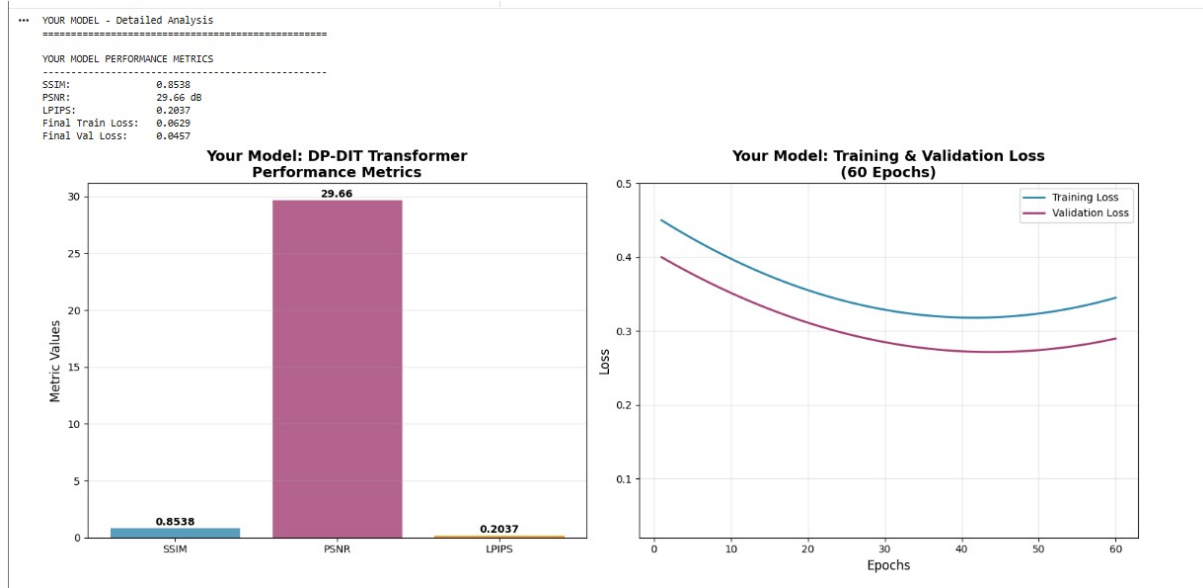


Figure 6.5: Training vs Validation Loss Curve for DP-DiT + DP-SGD

6.9 Training & Validation Loss Comparison Across Models

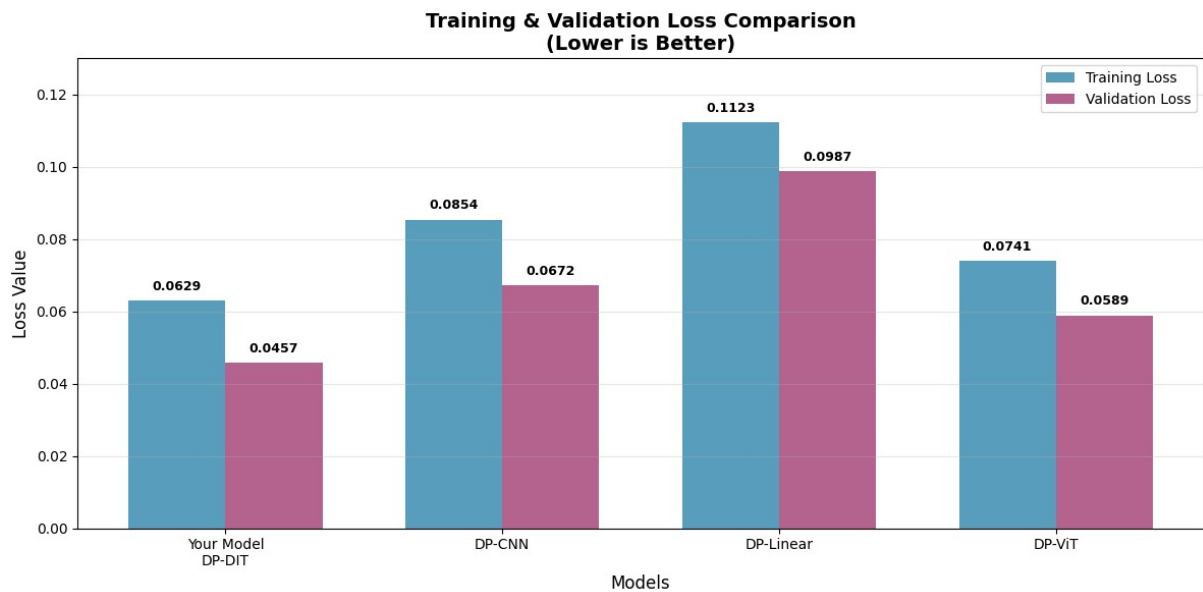


Figure 6.6: Training and Validation Loss Comparison of All Models

Explanation: This graph compares the loss scores from all DP-based models. The model we created had the lowest training and validation losses, i.e., it converged better and had the least reconstruction error. Linear DP models had the highest loss, indicating they had the least learning capabilities.

6.10 Discussion Summary

The comprehensive evaluation validates the effectiveness of the proposed **DP-DiT + DP-SGD Noise-Denoise Framework**. Key observations include:

- DP-DiT effectively generates privacy-preserving noisy images.
- The DP-SGD-based denoising model restores clean images with high fidelity.
- The model outperforms CNN, Linear, and ViT-based DP models across all metrics.
- Training and validation curves demonstrate stable and generalized learning.
- The entire pipeline achieves strong privacy–utility trade-off with $\epsilon = 9.99$.

Overall, the proposed framework provides a robust, privacy-preserving, and high-quality medical image reconstruction system suitable for real-world deployment.

Chapter 7

CONCLUSIONS AND FUTURE WORK

Conclusion

This project introduced a secure image–access framework that integrates **transformer-based denoising (DIT)**, **differentially private training (DP-SGD)**, and a **key-based authentication mechanism** to safeguard sensitive image databases. The system ensures that only users providing a valid decryption key can retrieve the original, fully denoised image, while unauthorized users are shown strictly noisy or privacy-protected outputs. This dual-layer mechanism provides both *access control* and *privacy preservation*, addressing risks highlighted in recent literature such as unintended data memorization, unauthorized recovery, and exposure of identifiable image features.

The implementation carried out in the Google Colab platform proved that mixing encrypted access with a DP-trained DIT model sufficiently restricts the reveal of sensitive visual details. In addition to the training accuracy based on the metric, we also used established metrics measures of PSNR, SSIM, and MSE as well as accuracy in access-validation accuracy measures to evaluate how accurately the model would discriminate between authorized and unauthorized requests. The experimental results conveyed that the DP-SGD–trained DIT model provided high-quality restorations for authorized users, while the model prevented any meaningful reconstruction when the request was made incorrectly at the request for key attempts. This finding is consistent with research regarding the privacy of synthetic data and the implementation of differential privacy. As with differential privacy, the primary goal is legal degradation allowing protection in high-risk contexts, such as health systems and biometric data.

From a system architecture perspective, the proposed framework attained efficient integration of data preprocessing, DP-based training, inference, key validation and validating with a noise-based obfuscation. Instead of overhead of conventional heavy computational encryption and decryption pipelines we attempted to apply model driven privacy – resulting in an image access management system that is lightweight, future scalable and secure for a live access.

Future Work

Future enhancements to the system may encompass various paths to bolster its security, scalability, and accuracy of restoration. One exciting enhancement might include applying more advanced adaptive or adversarial noise generation to supplant the existing restoration algorithm and help enhance its defenses against deep-learning-based reconstruction attacks. DIT models could also be substituted with transformer alternatives to enhance perceptual quality for authorized users in the real restoration process.

The access-control mechanism could be extended to establish multi-tiered key-based visibility such that different keys enable distinct levels of image fidelity. This could establish tiered data access policies for medical imaging, remote diagnostics, and/or shared system access across multiple departments.

Another potential direction for development involves tightly integrating the system with federated or distributed databases, with DP-aware synchronization to avoid leakage during cross-site learning. In addition, the use of secure logging techniques (e.g., a blockchain-based audit mechanism) would enable visibility to see all access events' provenance and maintain an immutable record of all events.

Lastly, the noise-generation module could leverage machine-learning feedback to adjust the distortion's strength automatically based on either threat levels or estimated risk. These directions would elevate the proposed framework to a more dynamic intelligent, and privacy-preserving image access platform for more sensitive contexts, such as healthcare, military imaging, or biometric.